

Week 3 Practice Problem: Conditions and Decision Making

Understanding Conditions and Selection Statements

Part A: Evaluate Conditions

For each condition below, determine whether the condition evaluates to **True** or **False**.

Assume:

```
int x = 5;  
int y = 10;
```

1. $x < y$ $5 < 10$
True
2. $x > y$ $5 > 10$
False
3. $x \leq 5$ $5 \leq 5$
True
4. $y == 10$ $10 == 10$
True

Part B: Evaluate Logical Expressions (Show work)

Assume:

```
int age = 25;
```

1. $\overset{25}{age} \geq 18 \checkmark \ \&\& \ \overset{25}{age} \leq 65 \checkmark$
True
2. $age < 18 \times \ || \ age > 65 \times$
False
3. $!(age \times \leq 18) \checkmark$
True
4. $age == 25 \checkmark \ \&\& \ age > 30 \times$
False

Part C: Predict Program Output

Determine the output.

Example 1

```
int score = 90;
if (score >= 70)
{
    printf("Pass");
}
```

Output:

Pass

Example 2

```
int score = 65;
if (score >= 70)
{
    printf("Pass");
}
else
{
    printf("Fail");
}
```

Output:

Fail

Example 3

```
int x = 10;
if (x > 0)
{
    printf("Positive");
}
else if (x < 0)
{
    printf("Negative");
}
else
{
    printf("Zero");
}
```

Output:

Positive

Part D: Trace the Decision

For each value of temperature, identify which message prints.

```
if (temperature < 32)
{
    printf("Freezing");
}
else if (temperature < 60)
{
    printf("Cool");
}
else
{
    printf("Warm");
}
```

Case 1

temperature = 20;

Output: *Freezing*

Case 2

temperature = 55;

Output: *Cool*

Case 3

temperature = 80;

Output: *Warm*

Part E: Write a Condition

Write a C condition for each situation.

1. A value `score` is at least 60.

$(score \geq 60)$

2. A value `age` is between 18 and 65 inclusive.

$(age \geq 18 \ \&\& \ age \leq 65)$

3. A character `grade` is either 'A' or 'B'.

$(grade == 'A' \ || \ grade == 'B')$

4. A number `num` is not equal to 0.

$(num \neq 0)$